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THEATRE ARTS

SET DESIGN

The Model Box, the Elevation and the Build

A ZikShare teaching unit, written for this shelf

From white card to workshop drawing: how to build a scale model, read a ground plan and elevation, and hand a set to a carpenter without losing the design.

WHAT THIS UNIT COVERS

- scale
- the model box
- elevations
- materials
- the build meeting

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Scale is a promise

1:25 means one centimetre on the model is twenty-five centimetres in the theatre; 1:50 is the plan scale most workshops prefer for a full stage. Choose one and stay on it, because a model built at one scale and a plan drawn at another will produce a set that does not fit.

Build the model box first: a simple card box open at the front and top, with the stage floor at the right height and the proscenium opening cut to size. Everything you put in it is judged against the box, which is the point: the box is the theatre.

The model in white

Build the model in white card or foam board before you colour it. In white you see proportion, sightlines and the arrangement of masses; colour hides bad proportion. Photograph it from the front, from both sides and from above, and look at the photographs as if you were an audience member.

Put scale figures in it. A one-centimetre figure on a 1:25 model is a 25 cm doll; two or three of them will tell you immediately whether a doorway is too low or a table too tall.

The elevation and the build drawing

An elevation shows each wall flat, with dimensions and the position of every opening and fixing. A carpenter builds from the elevation; the designer builds from the model. The two must agree, which means the elevation is drawn from the model, not from memory.

Dimension everything that matters: height, width, the distance from the floor to a shelf, the depth of a step. Write the materials on the drawing: 18 mm ply, 50 x 75 mm timber, 3 mm hardboard. If the workshop has to guess, it will guess cheaply.

The build meeting

Before building, meet the stage manager, the carpenter, the painter and the lighting designer together, with the model, the plan, the elevations and the schedule. Agree four things: what gets built first, what can be cut if time runs out, who paints it, and when the lighting designer needs the set finished. Write it down and circulate it.

Then go to the workshop every day during the build. A design is not finished when it is drawn; it is finished when it is standing on the stage and lit.

Finishing and photographing the model

A white model is for thinking; a finished model is for showing. Finish with texture: paint the floor, paper the walls, add the fabric of the curtains at scale, and use a matt finish because gloss photographs as plastic. Card edges should be clean; a sloppy model reads as a sloppy design, whatever the idea.

Light the model before you photograph it. A desk lamp at the angle of the stage light will show you whether the set has depth or reads flat. Photograph from the height of the audience's eye, not from above: the plan view is for the workshop, the eye view is for the director.

Keep the model after the production. Store it flat, photograph it from four angles, and file the photographs with the

drawings. Models are the department's memory, and the next production of the same play will begin with them.

When the model meets the theatre

A model is built to the theatre's dimensions, not to an ideal stage. Measure the venue: the proscenium opening, the depth to the back wall, the height to the grid, the width of the wings, the position of the doors the audience uses. A set designed for a stage two metres deeper than the one you have will be cut in the week before opening, badly.

Take the model to the venue and stand it in the actual space before the build. The perspective of a model flatters; the theatre does not. Photograph the model in the venue, from the seats, and adjust before anyone buys timber.

Exercises

1. Build a 1:25 white model of one room from a play you know, including two doors and one window.
2. Draw the elevations for that room with dimensions and materials.
3. Photograph the model from the worst seat in your theatre and revise one thing you can no longer see.

Where to read more

Darwin Reid Payne, Theory and Craft of the Scenic Model (1975)

Lynn Pecktal, Designing and Drawing for the Theatre (1995)

J. Michael Gillette, Theatrical Design and Production (2012)

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